

BUTUAN, Philippines

WMO: 987520

Lat: 8.9472N

Lon: 125.4823E

Elev: 151

StdP: 14.62

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 72.4	(c) 73.5	(d) 69.6	(e) 109.6	(f) 75.9	(g) 70.8	(h) 114.2	(i) 76.7	(j) 21.3	(k) 78.7	(l) 16.7	(m) 81.1	(n) 1.6	(o) 140	(p) 0.327

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 5	(b) 13.7	(c) 93.6	(d) 81.5	(e) 92.6	(f) 81.3	(g) 91.8	(h) 81.1	(i) 83.3	(j) 90.6	(k) 82.7	(l) 90.1	(m) 82.2	(n) 89.5	(o) 8.4	(p) 340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.1	162.5	88.1	80.6	159.9	87.7	80.1	156.9	87.1	47.3	90.9	46.6	90.5	46.1	90.0	87.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 15.2	(b) 13.0	(c) 11.3	(e) 68.3	(f) 96.7	(g) 4.0	(h) 1.6	(i) 65.4	(j) 97.8	(k) 63.1	(l) 98.7	(m) 60.9	(n) 99.6	(o) 58.0	(p) 100.8		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.5	80.0	80.5	81.6	83.0	84.4	83.7	83.3	83.6	83.4	83.1	81.8	81.4
	DBStd	2.26	2.25	2.10	1.96	1.86	1.63	1.78	1.67	1.64	1.56	1.51	1.87	1.96
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11860	929	853	980	991	1066	1012	1032	1040	1001	1027	955	974
	CDD65	6385	464	433	515	541	601	562	567	575	551	562	505	509
	CDH74	68856	3960	3963	5230	6177	7296	6482	6500	6643	6296	6260	5109	4940
	CDH80	25656	1037	1195	1812	2471	3173	2636	2576	2698	2502	2378	1673	1505
Wind	WSAvg	3.9	3.4	3.6	3.7	3.7	4.1	4.0	4.1	4.5	4.4	4.1	3.3	3.5
Precipitation	PrecAvg	89.5	13.0	9.4	6.1	4.7	5.1	7.7	6.7	5.5	5.9	6.7	8.7	10.1
	PrecMax	136.2	31.7	20.0	14.2	11.8	9.6	17.5	14.0	9.0	15.7	14.4	18.0	24.8
	PrecMin	55.4	1.3	0.7	0.2	0.4	2.9	2.6	2.6	1.9	1.5	2.6	3.3	3.2
	PrecStd	18.5	7.1	5.4	3.3	3.0	1.7	3.4	2.8	2.0	2.8	2.6	4.3	5.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.6	91.1	92.6	94.1	94.8	94.3	93.6	93.6	93.4	93.2	91.9	91.7
		MCWB	80.0	80.0	79.9	81.3	81.5	82.4	81.6	81.2	81.9	81.2	81.2	81.0
	2%	DB	88.7	89.3	90.7	92.4	93.3	92.6	92.0	92.2	91.8	91.5	90.2	89.9
		MCWB	79.6	79.4	79.9	80.9	81.7	81.8	81.2	81.0	81.3	81.0	80.8	80.4
	5%	DB	87.2	88.0	89.3	91.0	92.1	91.3	90.6	90.9	90.6	90.3	89.0	88.6
		MCWB	79.3	79.2	79.7	80.6	81.5	81.5	80.9	80.7	80.9	80.7	80.4	80.2
	10%	DB	85.5	86.5	88.0	89.7	90.9	90.0	89.4	89.7	89.4	89.0	87.6	87.0
		MCWB	78.9	78.8	79.3	80.4	81.3	81.1	80.6	80.4	80.7	80.5	80.1	79.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.9	81.9	82.6	83.4	83.7	83.7	83.2	83.1	83.6	83.3	82.6	82.6
		MCDB	88.0	88.2	89.4	90.7	91.3	91.4	90.4	90.8	91.0	90.0	89.2	89.1
	2%	WB	80.8	80.8	81.5	82.4	82.8	82.7	82.3	82.1	82.4	82.2	81.8	81.6
		MCDB	86.5	87.1	88.2	90.0	90.6	90.4	89.9	89.6	89.9	89.1	88.1	87.8
	5%	WB	80.0	80.0	80.7	81.6	82.3	81.9	81.6	81.5	81.6	81.6	81.1	80.8
		MCDB	85.3	86.1	87.3	89.0	90.0	89.4	89.0	88.9	88.9	88.3	87.2	86.8
	10%	WB	79.2	79.3	79.9	80.9	81.7	81.4	80.9	80.9	80.9	80.9	80.5	80.1
		MCDB	84.2	85.1	86.1	88.1	89.1	88.6	88.0	88.1	87.9	87.5	86.4	85.6
Mean Daily Temperature Range	5% DB	MDBR	10.2	11.7	12.6	13.4	13.7	13.2	12.9	13.1	13.1	12.8	11.9	11.3
		MCDBR	13.6	14.4	14.9	15.3	15.3	14.9	14.7	14.8	14.7	14.5	14.1	13.9
		MCWBR	6.4	6.7	6.8	6.7	6.4	6.4	6.4	6.2	6.3	6.2	6.4	6.3
	5% WB	MCDBR	12.2	12.9	13.7	14.3	14.3	14.2	13.9	14.0	14.3	13.6	13.2	12.8
		MCWBR	6.3	6.6	6.8	6.8	6.4	6.6	6.7	6.5	6.8	6.5	6.5	6.4
Clear-Sky Solar Irradiance	taub	0.438	0.426	0.420	0.420	0.420	0.412	0.417	0.442	0.438	0.444	0.421	0.429	
	taud	2.385	2.403	2.432	2.427	2.444	2.475	2.455	2.367	2.373	2.356	2.441	2.414	
	Ebn at Noon	277	284	286	282	277	276	275	272	277	275	280	277	
	Edh at Noon	38	39	38	38	36	35	36	40	40	40	36	37	
All-Sky Solar Radiation	RadAvg	1164	1392	1610	1741	1711	1586	1552	1621	1602	1515	1325	1226	
	RadStd	162	154	136	127	120	101	101	101	126	108	124	110	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page